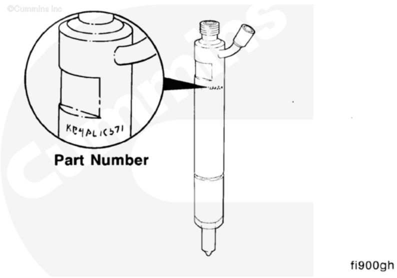
Trainings

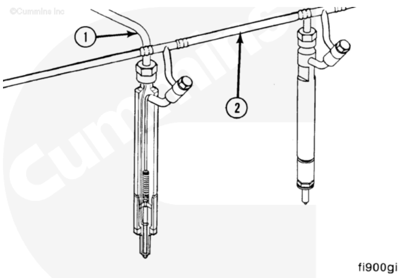
General Information

All engines use 7-mm closed nozzles, hole-type injectors. However, the injectors can have different part numbers for different engine ratings. The last four digits of the Cummins part number is used to identify the injectors.

Use **only** the specified injector for the engine.



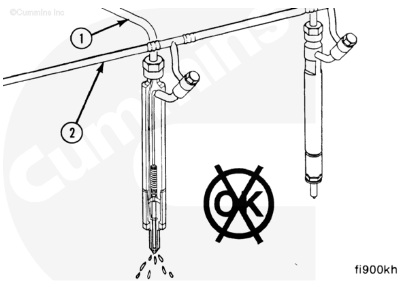
During the injection cycle, fuel under high pressure from the fuel injection pump rises to the operating (pop-off) pressure that causes the needle valve in the injector to lift. Fuel is then injected into the cylinder. A shimmed spring is used to force the needle valve closed as the injection pressure drops below the pop-off pressure. The shimmed spring seals off the nozzle after injection.



1. High-Pressure Fuel Line
2. Fuel Drain Manifold.

Failure of the needle valve to lift and close at the correct time, or the needle valve sticks open, can cause the engine to misfire and produce low power.

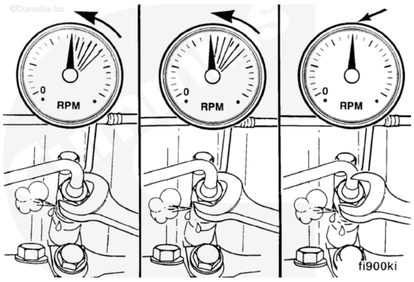
Fuel leaking from the open nozzle can cause:



1. Fuel knock
2. Poor performance
3. Smoke
4. Poor fuel economy
5. Rough-running engine.

Initial Check

To find which cylinder is misfiring, operate the engine and loosen the fuel line nut at one injector, and listen for a change in engine speed.

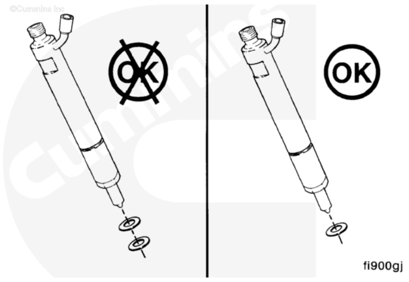


A drop in engine speed indicates the injector was delivering fuel to the cylinder.

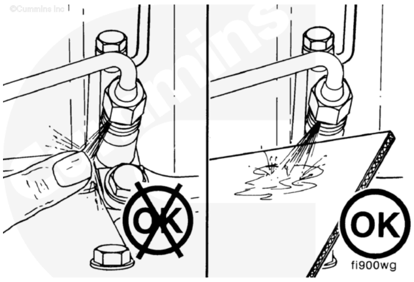
Inspect each cylinder until the malfunctioning injector is found.

Make sure to tighten the fuel line nut before proceeding to the next injector.

Remove the malfunctioning injector to test it.



Check for an extra copper sealing washer on the injector.



⚠ WARNING ⚠

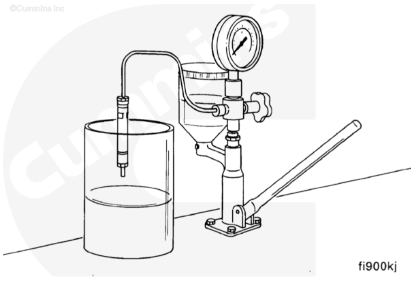
While testing injectors, keep hands and body parts away from the injector nozzle. Fuel coming from the injector is under extreme pressure and can cause serious injury by penetrating the skin.

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

If the engine continues to misfire, use cardboard to check for fuel leaks in the high-pressure lines. With the engine running, move the cardboard over the fuel lines, and look for fuel spray on the cardboard. Fuel leaks can cause poor engine performance. Also, check for a defective delivery valve that lets the fuel drain back into the injection pump.

Carbon buildup in the orifices in the nozzle can also cause low power from the engine. Remove the injectors and check the spray pattern, or replace the injectors.



Preparatory Steps

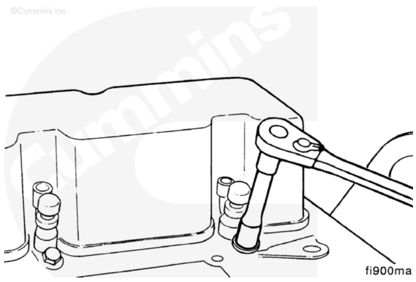
Thoroughly clean around the injectors.

Disconnect the high-pressure injector supply lines; refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html).

Disconnect the fuel drain manifold; refer to Procedure 006-021 (/qs3/pubsys2/xml/en/procedures/41/41-006-021-tr.html).

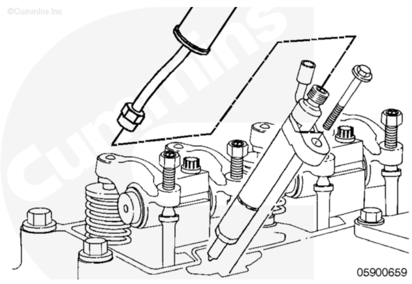
Remove

Remove the injector hold-down clamp.



Use injector puller, Part Number 3823276 or 3164706, to remove the injectors.

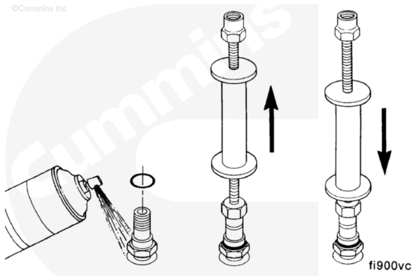
Remove the injectors.



To remove some injectors, it will probably be necessary to:

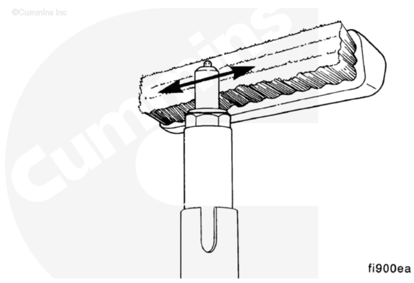
- Tap the injector with the injector puller

- Work the injector up and down.

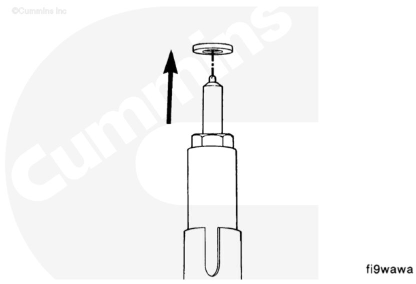


Disassemble

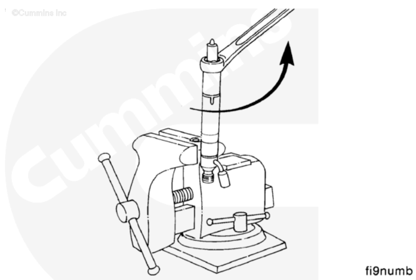
Clean the carbon residue from the injector nozzle, using a brass wire brush and a piece of hardwood dipped in test oil.



Remove and discard the injector seals.

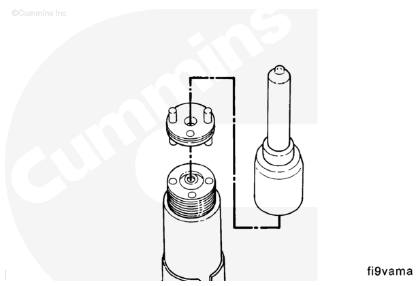


Clamp the injector hold-down clamp in a soft-jawed vise. Remove the injector nozzle retaining nut.



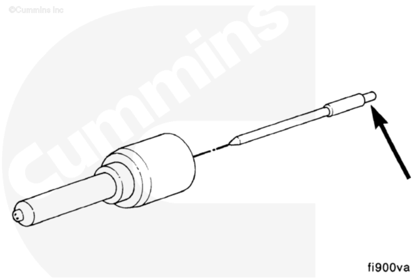
Remove the nozzle needle valve and intermediate plate.

To avoid damage, place the injector nozzle and needle valve in a suitable bath of clean test oil.

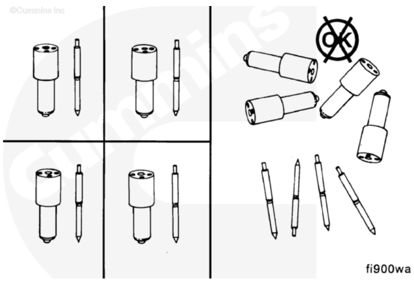


⚠ CAUTION ⚠

Hold the needle valve by the stem only. Secretions from the skin will corrode the finely lapped surfaces.

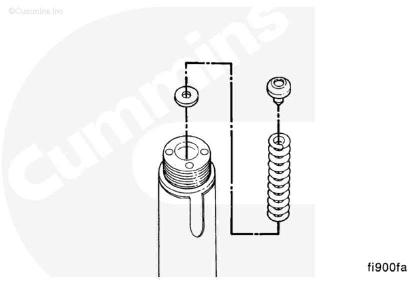


The needle valve and nozzle tip are precisely matched for fit. The parts **must not** be intermixed.

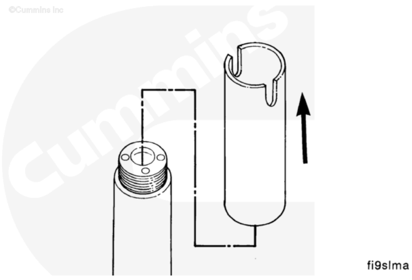


Remove the nozzle holder from the vise.

Remove the pressure spindle, pressure spring, and shims.



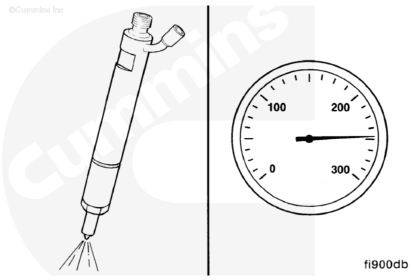
Remove and discard the injector sealing sleeve.



Test

⚠ WARNING ⚠

While testing injectors, keep hands and body parts away from the injector nozzle. Fuel coming from the injector is under extreme pressure and can cause serious injury by penetrating the skin.



⚠ WARNING ⚠

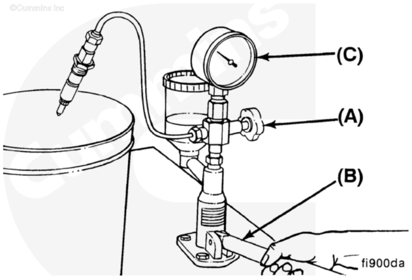
The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

Injector Opening Pressure Test

All nozzles **must** be tested for opening pressure, chatter, and spray pattern.

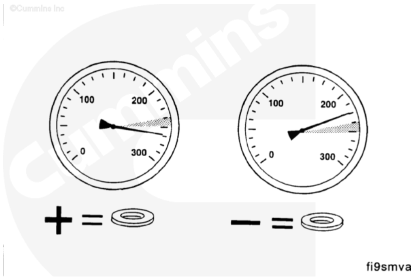
Inspect the injector opening pressure using injector nozzle tester, Part Number 3376946:

A. Open valve.
B. Operate the lever at one stroke per second.
C. Read the pressure indicated when the injector spray begins.



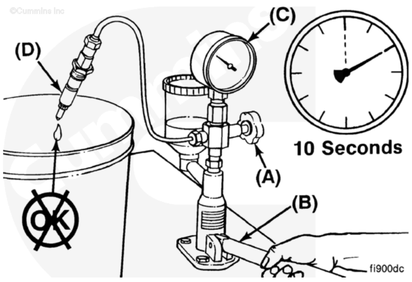
If the opening pressure does **not** meet specifications:

1. Add shim(s) to increase pressure.
2. Remove shim(s) to decrease pressure.



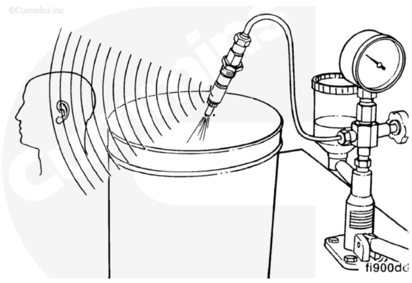
Leakage Test

1. Open valve (A).
2. Operate the lever (B) to maintain a pressure of 20 bar [290 psi] below opening pressure (C).
3. No drops can fall from the tip (D) within 10 seconds.



Chatter Test

The chatter test indicates the ability of the needle valve to move freely and atomize the fuel correctly. A sound will be heard as the valve rapidly opens and closes. A well-optimized spray pattern will be seen.

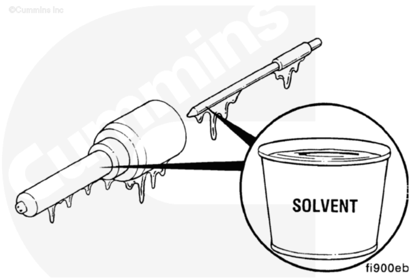


Used nozzles can **not** be evaluated for chatter at lower speeds. A used nozzle can generally be used if it passes the leakage test.

Clean and Inspect for Reuse

⚠ WARNING ⚠

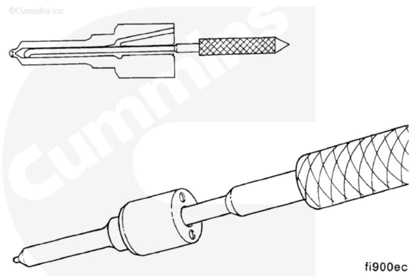
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Rinse the nozzle bodies and needle valves in solvent to flush and remove thoroughly and completely all varnish and carbon deposits.

⚠ CAUTION ⚠

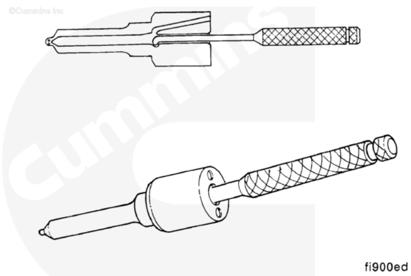
Never use emery paper or any other metal scraper to clean the nozzle. The parts can be damaged.



Dip the nozzle seat in clean test oil, and use nozzle cleaning kit, Part Number 3376947, to clean. Polish the needle seat with the piece of hardwood dipped in the test oil.

⚠ WARNING ⚠

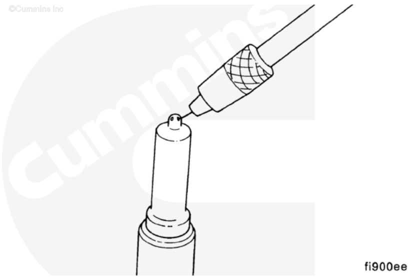
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



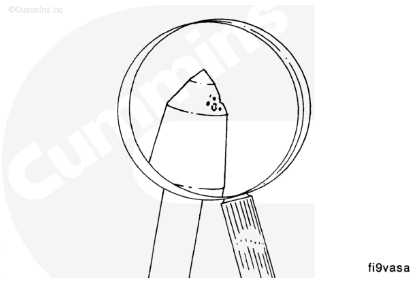
Clean the interior ring groove of the nozzle with the scraper as shown. Rinse in solvent to remove all dirt and carbon residue and dip in clean test oil.

Clean the spray holes as shown with the appropriate-size cleaning needle.

Remove burned-on combustion deposits on all nozzles with a commercially available cleaner. Rinse all parts in clean test oil.



Clean the needle valve tip with a brass brush. Inspect for rough surfaces or erosion. The pressure shoulder will normally have a rough machined appearance.



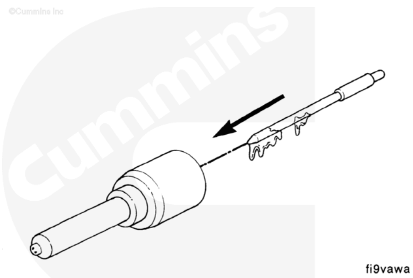
Deteriorated needle valves **must** be replaced as a matched unit with their compatible nozzle body.

Inspect the injector. Inspect the o-ring for damage. Inspect for burrs on the inlet to the injector. Check the nozzle holes for any signs of damage, such as hole erosion or hole plugging. Also, check the nozzle color for sign of overheating. Overheating will cause the nozzle to turn a dark yellow and tan or blue color, depending on the temperature of the overheat.

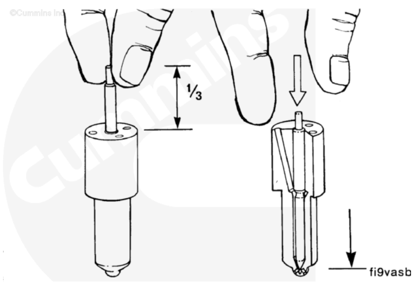
Inspect for rough surfaces or erosion. The pressure shoulder will normally have a rough machined appearance.

Inspect the injector bore for old sealing washers.

Dip the needle valve in clean test oil, and insert the needle valve all the way into the nozzle body.



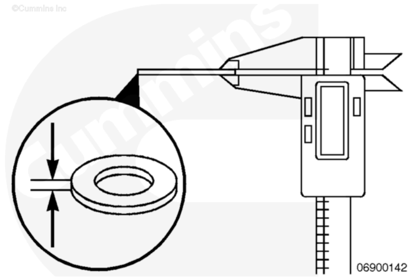
Pull the needle valve one-third of the way out of the nozzle body. With the needle valve in the vertical position, the needle valve **must** slide all the way back into the nozzle body under its own weight.



If the nozzle does **not** pass this test, clean and test it an additional time.

Any needle valve and nozzle body assembly that does **not** pass this test **must** be replaced.

Verify that the injector sealing washer is the correct thickness. The incorrect sealing washer can cause high-pressure fuel leaks, and/or performance problems due to incorrect injector protrusion.

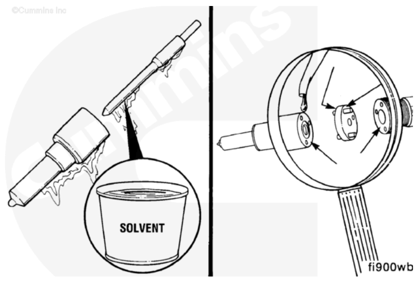


Measurements		
	mm	in
Injector Sealing Washer	1.5	0.06

Assemble

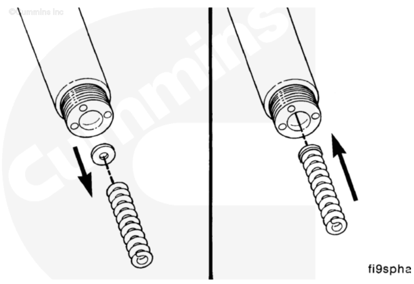
Make sure all mating surfaces and pressure faces are thoroughly cleaned and lubricated with test oil before assembly.

New nozzles **must** be cleaned and lubricated before assembly.

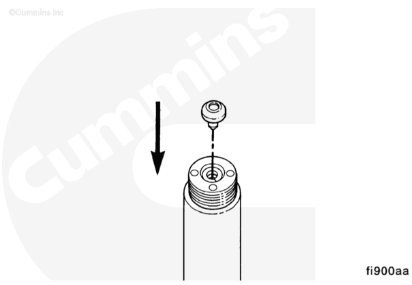


Install the shim(s) and pressure spring.

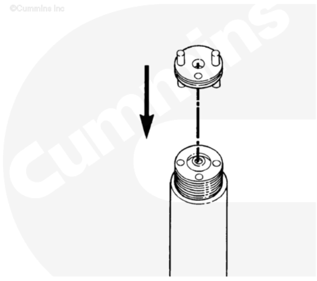
Install the same thickness of shim(s) that were removed in disassembly. Use the pressure spring to make sure the shim(s) are installed flat.



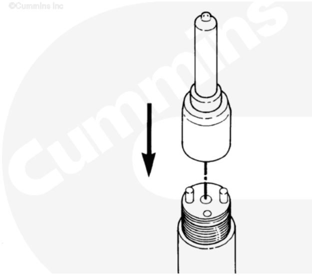
Install the spindle.



Install the intermediate plate.


fi9plaa

Install the needle valve and nozzle assembly.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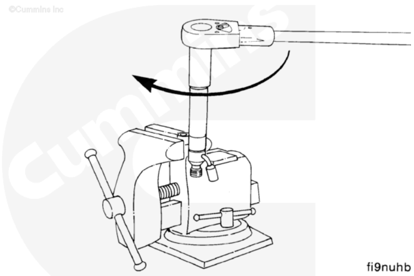
Install and tighten the nozzle retaining nut.

Torque Value: 30 n•m [22 ft-lb]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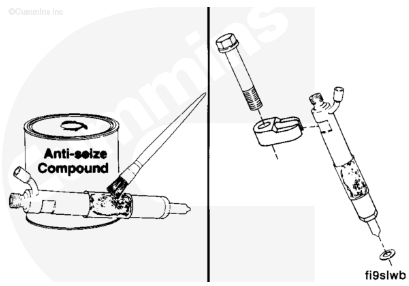
Install

Lubricate the sealing lips of the sleeve with anti-seize lubricant, Part Number 3824879. Assemble the injector, sealing sleeve, a new copper sealing washer, and the hold-down clamp.

Use **only** one washer.

Service Tip
A light coat of clean engine oil between the washer and injector can help hold the washer in place during installation.



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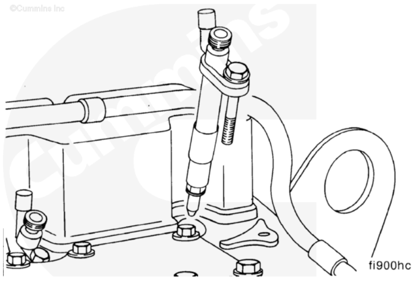
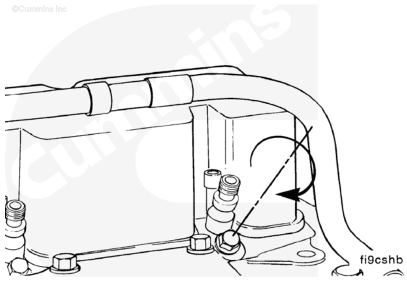
Install the injector assembly into the injector bore. The injector drain connection **must** be toward the valve cover.



Install and tighten the hold-down capscrew.

Torque Value: 28 n•m [21 ft-lb]



Finishing Steps

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

Install the fuel drain lines. Refer to Procedure 006-021 (/qs3/pubsys2/xml/en/procedures/41/41-006-021-tr.html).

Install the high pressure fuel lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html).

Last Modified: 07-Dec-2004